

LLM PROMPTS AND ANSWERS

Panel Data Econometrics Homework: Replication

Mertzanis (2024) — Central Bank Policies and Green Bond Issuance on a Global Scale
Energy Economics, 133, 107541

Session dates: June 5–8, 2026

LLM used: Claude (Anthropic)

Document generated: June 8, 2026

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About this document

This document contains a complete record of LLM (Claude, Anthropic) prompts and responses used to complete the Panel Data Econometrics replication homework. It covers three working sessions between June 5 and June 8, 2026, and documents all code generation, data exploration, econometric decisions, and translation steps.

The document is organized chronologically by session. Each exchange is presented with a USER block (the student's request or information provided) followed by an ASSISTANT block (Claude's response, including all code generated).

Total sessions: 5 parts. Primary language: Python (with Stata and R translations).

Part 1 — Initial Project Setup (June 5, 2026)

USER

I am a student working on a Panel Data Econometrics replication project with a 2-day deadline.

I chose to replicate:

Mertzanis, C. (2024). Central bank policies and green bond issuance on a global scale. *Energy Economics*, 133, 107541.

Data: Mendeley Data, doi: 10.17632/xb2m4m8h24.1

Please provide:

1. A detailed project plan for the full replication workflow
2. A recommended folder structure
3. A first version of the Python workflow
4. Analysis of the requirements files already in the workspace

Files already in workspace:

- panel_replication_requirements.md (full homework assignment specification)
- ftd_panel_replication_list_of_papers.md (list of 13 papers to replicate)

Key requirements from the assignment:

- Minimum 3 consecutive observations of Y per country (panel selection rule)
- Compute: between, one-way within, TWFE, first-difference transformations
- Variance decomposition (% within vs between for each variable)
- Descriptive statistics + plots (histogram, KDE, normal, bivariate)
- Five estimators: Between, One-Way FE, Mundlak RE, TWFE, First Differences
- Optional: Anderson-Hsiao dynamic panel if autocorrelation of $\Delta Y > 0.1$
- Submit: Python code, Stata translation, report (Word/LaTeX), LLM prompt log

ASSISTANT

I read both requirement files in full and proceeded to build the complete project structure.

Actions taken:

1. Created the folder structure:

```
data/raw/      data/processed/  scripts/
outputs/tables/ outputs/figures/  outputs/logs/
report/        paper/
```

2. Created PROJECT_PLAN.md with a 2-day schedule and 12-step plan.

3. Created README.md with quick-start instructions.

4. Created 8 Python scripts (initial version with placeholder variable names):

```
config.py
01_load_explore_data.py
02_clean_panel_sample.py
03_panel_transformations.py
04_descriptive_statistics.py
05_panel_estimations.py
06_dynamic_panel.py
main_workflow.py
```

The 12-step plan covers:

- Step 0: Environment setup and package installation
 - Step 1: Paper reading and variable identification
 - Step 2: Load and inspect raw data
 - Step 3: Panel sample selection (≥ 3 consecutive obs rule)
 - Step 4: Panel structure description (attrition, T distribution)
 - Step 5: Variable classification (between vs within variance)
 - Step 6: Distribution plots (between + one-way within)
 - Step 7: First differences and TWFE transformations
 - Step 8: Descriptive statistics tables (all 4 transforms)
 - Step 9: Panel regressions (Between, FE, Mundlak RE, TWFE, FD)
 - Step 10: Anderson-Hsiao dynamic panel (optional)
 - Step 11: Time-invariant variables (Hausman-Taylor, interaction shortcut)
 - Step 12: Output organization and report
-

Part 2 — Data Downloaded and Environment Issue (June 5, 2026)

USER

I did step 1 — downloaded the data from Mendeley.

Files now in data/raw/:

- data.xlsx (the actual dataset)
- Stata_code.docx (the authors' original Stata replication code)

ASSISTANT

Issue detected: the openpyxl library on the user's Anaconda installation is outdated (incompatible with numpy ≥ 1.20 due to deprecated np.float alias).

Attempted fix: pip install --upgrade openpyxl

However, Python was not accessible from the terminal in this environment.

Before rebuilding the scripts with actual variable names, I asked two clarifying questions:

1. What are the exact column names in data.xlsx?
2. What is the content of Stata_code.docx (the authors' original code)?

USER

I am not allowed to run code here — I will run on another machine.

Please provide me with the full constructed project, which is maximum done.

Leave me only the running actions with the plan for further steps.

Part 3 — Full Rebuild with Actual Data (June 5, 2026)

3.1 — Variable information provided by student

USER

Excel column headers (row 1 of data.xlsx):

country | country-numeric | year | iso3 | income level, 2-levels |
 1 if country=China, 0 otherwise |
 log value of total green bond issuance value |
 log value of government green bond issuance value |
 log value of corporate green bond issuance value |
 Environmental Taxes (%GDP) | GDP growth (annual %) [NY.GDP.MKTP.KD.ZG] |
 bitech finance as a percent of GDP |
 Notre Dame-Global Adaptation Index to climate disruptions |
 Moody's rating, numeric 4-1 | Total equity risk premium, Damodaran |
 Country risk premium, Damodaran | 1 if member of EU27 | 1 if member of OECD |
 IMF: financial institutions efficiency | square value of CB mandate |
 CB mandate - rowtotal | square value of CB enforcement |
 CB enforcement - rowtotal | square value of CB control of syst risk tools |
 CB control sys risk tool - rowtotal | world uncertainty index |
 log of geopolitical risk index | Official Core Consumer Price Inflation, IMF |
 Share of primary energy consumption that comes from oil |
 1st principal component of value added: agriculture, manufacturing, industry, services |
 Hierarchy-Schwartz-2007 | Openness-Schmitt et al.-2007 | Purity-Atari et al.-2020

Stata code from Stata_code.docx (authors' original replication code):

```
global xvar cbmnd cbenf cbsys infl rating fie gdpg dum9 dum12
```

TABLE 1: tabstat bndgrn cbmnd cbenf cbsys by(country1)

TABLE 2: summarize bndgrn \$xvar, listwise detail

TABLE 3: correlate bndgrn \$xvar + collin (VIF)

TABLE 4: reg bndgrn \$xvar [i.year] [i.country1], cluster(country1) -- 4 specs

TABLE 5: robustness -- gov bonds, corp bonds, devlev, no China, non-linear

TABLE 6: ivreg2 -- each CB var instrumented by hierarchy/purity/openness

TABLE 7: reg bndgrn \$xvar + extra controls (risk premiums, energy, geopolitical)

TABLE 8: domin (dominance analysis of R2)

TABLE 9: reg bndgrn \$xvar + interact_var + c.cbenf#c.interact_var

3.2 — Variable mapping and paper structure analysis

ASSISTANT

Based on the column headers and Stata code, here is the complete variable mapping (Excel long name → Stata short name):

log value of total green bond issuance value	→ bndgrn	(Y, main dependent)
--	----------	---------------------

log value of government green bond issuance value	→ x1l	(Y, alternative: gov bonds)
log value of corporate green bond issuance value	→ x2l	(Y, alternative: corp bonds)
CB mandate - rowtotal	→ cbmnd	(X1, key explanatory)
CB enforcement - rowtotal	→ cbenf	(X2, key explanatory)
CB control sys risk tool - rowtotal	→ cbsys	(X3, key explanatory)
Official Core Consumer Price Inflation, IMF	→ infl	(control)
Moody's rating, numeric 4-1	→ rating	(control)
IMF: financial institutions efficiency	→ fie	(control)
GDP growth (annual %) [NY.GDP.MKTP.KD.ZG]	→ gdpg	(control)
1 if member of EU27 invariant)	→ dum9	(control, time-invariant)
1 if member of OECD invariant)	→ dum12	(control, time-invariant)
square value of CB mandate	→ cbmndsq	(non-linear spec)
square value of CB enforcement	→ cbenfsq	(non-linear spec)
square value of CB control of syst risk tools	→ cbsyssq	(non-linear spec)
Total equity risk premium, Damodaran control)	→ riskeqt	(Table 7 extra)
Country risk premium, Damodaran control)	→ riskctr	(Table 7 extra)
Share of primary energy consumption from oil control)	→ enrgoil	(Table 7 extra)
Environmental Taxes (%GDP) control)	→ envtax	(Table 7 extra)
log of geopolitical risk index control)	→ gprlog	(Table 7 extra)
world uncertainty index control)	→ wui	(Table 7 extra)
Hierarchy-Schwartz-2007	→ hierarchy	(IV instrument)
Openness-Schmitt et al.-2007	→ openness	(IV instrument)
Purity-Atari et al.-2020	→ purity	(IV instrument)
bitech finance as a percent of GDP Table 9)	→ bigtech	(interaction var,
Notre Dame-Global Adaptation Index Table 9)	→ ngain	(interaction var,
1st principal component of value added... Table 9)	→ prodstr	(interaction var,
income level, 2-levels	→ devlev2	(subsample split)
1 if country=China, 0 otherwise	→ china	(subsample split)
country-numeric variable)	→ country_num	(cluster SE

Paper structure understood:

Y: bndgrn (log total green bond issuance, already log-transformed in raw data)

X: THREE CB policy variables simultaneously — cbmnd, cbenf, cbsys

Controls (xvar): infl, rating, fie, gdpg, dum9, dum12

Note: dum9 and dum12 are time-invariant → automatically absorbed by entity FE

IV logic: cultural/psychological values (Schwartz, Atari) used as instruments
for CB policy (time-invariant, so IV operates in cross-country dimension)

KEY DISCOVERY at this stage:

cbmnd, cbenf, and cbsys are TIME-INVARIANT — they have 0% within-country variance. This means entity fixed effects (FE) and first-difference (FD) estimators cannot identify their coefficients. This fundamental constraint shapes all

subsequent analysis and is the central methodological finding of the replication.

3.3 — Complete rewrite of all Python scripts

ASSISTANT

I performed a complete rewrite of all scripts using the actual variable names.
Below is a summary of each script's purpose and key implementation decisions.

config.py

```
# Complete variable configuration
RENAME_MAP = {
    'log value of total green bond issuance value': 'bndgrn',
    'log value of government green bond issuance value': 'x1l',
    'log value of corporate green bond issuance value': 'x2l',
    'CB mandate - rowtotal': 'cbmnd',
    'CB enforcement - rowtotal': 'cbenf',
    'CB control sys risk tool - rowtotal': 'cbsys',
    'Official Core Consumer Price Inflation, IMF': 'infl',
    'Moody's rating, numeric 4-1': 'rating',
    'IMF: financial institutions efficiency': 'fie',
    'GDP growth (annual %) [NY.GDP.MKTP.KD.ZG]': 'gdpg',
    '1 if member of EU27': 'dum9',
    '1 if member of OECD': 'dum12',
    # ... (full 33-column mapping)
}

Y_COL = 'bndgrn'
CB_COLS = ['cbmnd', 'cbenf', 'cbsys']
CONTROL_COLS = ['infl', 'rating', 'fie', 'gdpg', 'dum9', 'dum12']
XVAR_COLS = CB_COLS + CONTROL_COLS # matches paper's $xvar global
CB_SQ_COLS = ['cbmnds', 'cbenfs', 'cbsyss']
RISK_COLS = ['riskeq', 'riskctr']
ENERGY_COLS = ['enrgoil', 'envtax']
UNCERT_COLS = ['gprlog', 'wui']
IV_INSTRUMENTS = ['hierarchy', 'openness', 'purity']
INTERACT_VARS = ['prodstr', 'bigtech', 'ngain']
CLUST_COL = 'country_num' # cluster-robust SE by country
```

01_load_explore_data.py

```
# Load Excel, rename 33 columns using RENAME_MAP
# Produce paper Table 1: mean by country for bndgrn, cbmnd, cbenf, cbsys
# Outputs: t01a_obs_per_country.csv, t01b_missing_values.csv,
#          t01c_raw_descriptives.csv, t01d_means_by_country.csv
```

02_clean_panel_sample.py

```
# Panel selection rule: keep only countries with >=3 consecutive observations of
bndgrn
# Sort by country then year
# Detect consecutive runs; mark countries that never reach a run of 3+
# Replace isolated bndgrn values (runs < 3) with NaN
# Build sample selection table
# Check: are excluded countries systematically small/poor?
# Outputs: panel_clean.csv, t02a_sample_selection.csv,
#          t02b_obs_per_year.csv, t02c_T_distribution.csv,
#          fig01_obs_per_year.png
```


03_panel_transformations.py

```
# Compute for every variable:
#   Between:      x_i_bar = mean over time for each country
#   One-way within: x_it - x_i_bar
#   First differences: x_it - x_i,t-1 (NaN at every change of individual!)
#   TWFE balanced: x_it - x_i_bar - x_t_bar + x_grand_bar
#   TWFE unbalanced: regress within-Y on time dummies, take residuals
#                   (Wansbeek-Kapteijn 1989 method)
# Compute variance decomposition: overall, between, within, % within
# Classify variables: time-invariant (0%), mixed, individual-invariant (100%)
# Outputs: panel_transforms.csv, t03a_variance_decomp.csv,
#         t03b_variable_classification.csv, t03c_twfe_time_means.csv
```

04_descriptive_statistics.py

```
# Paper Table 2: summary stats on listwise regression sample (N=206)
# Paper Table 3: correlation matrix + VIF for all xvars (replicate Stata collin)
# Univariate distributions: histogram + KDE + fitted normal for Y and X
#   under all 4 transformations (between, within, TWFE, FD)
# Bivariate scatter: Y vs X with linear/quadratic/LOWESS fits
#   for each transformation
# Correlation matrices: between, within, TWFE, FD
# Heterogeneity tables: r(Y,X) by country sorted for TWFE and FD
# Boxplots by country for TWFE and FD
# Outputs: t04_paper_table2_summary_stats.csv, t04_paper_table3a_corr_matrix.csv,
#         t04_paper_table3b_vif.csv, t04a_univariate_all_transforms.csv,
#         t04b_corr_bet/wi/twfe/fd.csv, t04c_heterogeneity_twfe/fd.csv,
#         fig02-fig07 (distributions, bivariate, boxplots, corr heatmaps)
```

05_panel_estimations.py

```
# PART A – Paper replication:
#   Table 4: Pooled OLS, +year FE, +entity FE, TWFE (4 specifications)
#   Table 5: Robustness (gov bonds, corp bonds, developed, developing,
#               ex-China, non-linear with squared CB terms)
#   Table 7: Additional controls (risk premiums, energy, geopolitical)
#
# PART B – Homework additional estimators (not in paper):
#   Between OLS (linearmodels.BetweenOLS)
#   One-Way FE (linearmodels.PanelOLS, entity_effects=True)
#   Mundlak RE (RandomEffects + group means of X as extra regressors)
#   TWFE (linearmodels.PanelOLS, entity+time effects, drop_absorbed=True)
#   First Differences (linearmodels.FirstDifferenceOLS)
#
# Implementation notes:
#   - Paper specs use statsmodels.OLS with cov_type='cluster'
#   - dum9, dum12 automatically dropped in entity FE specs
#   - build_display_table() produces publication-ready format with stars
#     (***/**/*) and SE in parentheses
#
# Outputs: t05_table4_main.csv, t05_table5_robustness.csv,
#         t05_table7_add_controls.csv, t05_homework_estimators.csv
```

05b_iv_interactions.py (new script)

```
# Table 6 – IV/2SLS (operating in cross-country dimension):
#   Spec (1): cbmnd instrumented by hierarchy, purity
#   Spec (2): cbenf instrumented by hierarchy, openness
#   Spec (3): cbsys instrumented by hierarchy, openness
#   Uses linearmodels.iv.IV2SLS with fallback to manual 2SLS
#   Reports first-stage R2 and F-stat on excluded instruments
```

```
# Warning if first-stage R2 < 0.10 or F-stat < 10 (weak instruments)
#
# Table 9 – Interaction terms:
#   cbenf x prodstr, cbenf x bigtech, cbenf x ngain
#   Entity FE (country dummies), cluster SE
#   Marginal effect plots: d(bndgrn)/d(cbenf) as function of interact_var
#
# Outputs: t05b_table6_iv.csv, t05b_table6_first_stages.csv,
#          t05b_table9_interactions.csv, fig15_mfx_cbenf_*.png
```

06_dynamic_panel.py

```
# Anderson-Hsiao (1981) dynamic panel estimator
# Condition: only run if autocorrelation of DeltaY > 0.1
#
# Model:
#   DeltaY_it = beta_y * DeltaY_i,t-1
#               + beta_1 * DeltaX_it + beta_2 * DeltaX_i,t-1
#               + DeltaControls + DeltaAlpha_t + DeltaEpsilon_it
#
# Instruments: Y_i,t-2 and X_i,t-2 (levels, to instrument DeltaY_i,t-1)
# Steps:
#   1. Check autocorrelation condition
#   2. Compute instruments
#   3. OLS with lagged DeltaY
#   4. IV/2SLS using Anderson-Hsiao instruments
#   5. First-stage regression report (R2, F-stat)
#   6. Impulse response function for 4 periods:
#       t=1: beta_1
#       t=2: beta_y*beta_1 + beta_2
#       t=3: beta_y^2*beta_1 + beta_y*beta_2
#       t=4: beta_y^3*beta_1 + beta_y^2*beta_2
#   7. Long-run coefficient: beta_LT = (beta_1 + beta_2) / (1 - beta_y)
#
# Outputs: t06a-d_dynamic_*.csv, fig14_impulse_response.png
```

3.4 — Key econometric decisions

ASSISTANT

The following econometric decisions were made explicitly during the session:

1. Y variable: bndgrn is already log-transformed in the raw data.
No additional transformation needed.
2. Three X variables used simultaneously (not one):
cbmnd (mandate breadth), cbenf (enforcement power), cbsys (systemic risk control).
3. Time-invariant controls (dum9=EU27, dum12=OECD):
Included in Pooled OLS and year-FE specs.
Dropped (absorbed) in entity FE and TWFE specs automatically.
4. Cluster-robust SE: always clustered at country level (country_num).
5. TWFE implementation (two approaches for different contexts):
- In paper replication (05_panel_estimations.py): statsmodels OLS with explicit

country and year dummies.

- In homework estimators: `linearmodels PanelOLS` with `entity_effects=True`, `time_effects=True`, `drop_absorbed=True`.

6. IV instruments are time-invariant (cultural values from Schwartz/Atari).

Cannot be used with entity FE (they would be absorbed by country dummies).

IV specifications use year dummies only, no entity FE.

7. Mundlak RE: RandomEffects estimator + group means of all X as extra regressors.

This makes RE consistent even if entity effects are correlated with X

(equivalent to the Hausman test rejection correction).

8. First Differences: NaN inserted at every change of individual (country) in the sorted panel. This is critical to avoid computing the difference between the last observation of country i and the first observation of country i+1.

9. Anderson-Hsiao: only run if autocorrelation of $\Delta Y > 0.1$.

Instruments: $Y_{i,t-2}$ and $X_{i,t-2}$ (in levels, to instrument $\Delta Y_{i,t-1}$).

Part 4 — GitHub Setup Attempt (June 7–8, 2026)

USER

I want to add the project to GitHub.

Repository URL: <https://github.com/katerynavalenia/panel-replication-mertzanis2024.git>

ASSISTANT

Created a .gitignore file to exclude large auto-generated or proprietary files:

```
# .gitignore
data/raw/           # Mendeley data (not to be redistributed)
data/processed/     # auto-generated by scripts
outputs/            # auto-generated tables and figures
*.xlsx              # large binary files
*.pdf                # large binary files
*.docx              # large binary files
__pycache__/_       #
*.pyc                # Python cache
.venv/              # virtual environment
```

ASSISTANT

Issue encountered: Git is not installed on the machine (corporate Allianz laptop).

Diagnosis:

- C:\Program Files\Git\ does not exist
- git command not found in PATH
- winget and other package managers not available
- Machine has: Anaconda Python, VS Code, Microsoft Office, OneDrive

Solution offered: Upload via GitHub web interface using drag-and-drop.

Follow-up issue: GitHub web UI has a 100-file limit per upload and does not always accept full folder drag-and-drop reliably.

Recommended workaround: upload one subfolder at a time:

1. scripts/ folder (9 .py files)
2. Root files: README.md, PROJECT_PLAN.md, .gitignore, panel_replication_requirements.md, ftd_panel_replication_list_of_papers.md

Current status: GitHub repository created but not yet populated.

Part 5 — Submission Audit and Document Creation (June 8, 2026)

USER

Could you please verify this project in the folder and tell me if it is done fully according to the professor's requirements to get the highest grade, or what we should change or add?

Here is what we need to submit:

1. The original paper in PDF
2. The database in Excel format
3. The database formatted for Python
4. The Python code in runnable format
5. The code translated into Stata and R with LLM comments
6. A Word or TEX and PDF version of answers
7. A Word and PDF file with all LLM prompts and answers
8. A 'To-do list' of panel data steps

ASSISTANT

After inspecting all files in the project folder, here is the audit result:

PRESENT:

- | | |
|---|----|
| (1) paper_main.pdf | OK |
| (2) data/raw/data.xlsx | OK |
| (3) data/processed/panel_clean.csv | OK |
| (4) scripts/ folder + main_workflow.py | OK |
| (5) report/stata_translation.do + r_translation.R | OK |
| (6) report/panel_replication_report.docx + .tex | OK |

MISSING:

- (7) LLM prompts and answers document (Word + PDF) ABSENT
This is the current document being created.

- (8) PDF version of the answers report ABSENT
Need to compile .tex or export .docx to PDF.

CONTENT ISSUES TO FIX:

- Student name is blank in the report
- TWFE has only N=15 observations (balanced restriction too aggressive)
- Heterogeneity tables use cbmnd (time-invariant) instead of fie
- Section 12 (original estimations) needs genuine new analysis
- Dynamic panel output in report body needs trimming

USER

Create the LLM prompts document first.

ASSISTANT

Asked clarifying question: Do you have the actual LLM conversation logs?

User answered: Yes, I can paste/share the conversation.

User provided: LLM_conversation_log.txt (this conversation summary).

Proceeded to create this Word document from the provided log.

Appendix: Final Project File Structure

panel-replication-mertzanis2024-main/	
├─ paper_main.pdf	Original paper (Mertzanis 2024)
├─ data/	
│ └─ raw/	
│ └─ data.xlsx	Mendeley dataset (33 columns)
│ └─ Stata_code.docx	Authors' original Stata code
│ └─ processed/	
│ └─ panel_clean.csv	Panel after selection rule
│ └─ panel_raw.csv	Raw loaded data
│ └─ panel_transforms.csv	All 4 transformations
├─ scripts/	
│ └─ config.py	Variable names, paths, globals
│ └─ 01_load_explore_data.py	Load + rename + paper Table 1
│ └─ 02_clean_panel_sample.py	Panel selection (≥ 3 consec obs)
│ └─ 03_panel_transformations.py	Between, within, TWFE, FD
│ └─ 04_descriptive_statistics.py	Tables 2 & 3, all plots
│ └─ 05_panel_estimations.py	Tables 4, 5, 7 + homework estimators
│ └─ 05b_iv_interactions.py	Tables 6, 9 (IV, interactions)
│ └─ 06_dynamic_panel.py	Anderson-Hsiao dynamic panel
│ └─ main_workflow.py	Runs all 7 steps end-to-end
├─ outputs/	
│ └─ tables/	~25 CSV output tables
│ └─ figures/	~20 PNG figures
│ └─ logs/	7 step log files
├─ report/	
│ └─ panel_replication_report.docx	Main answers report (Word)
│ └─ panel_replication_report.tex	Main answers report (LaTeX)
│ └─ stata_translation.do	Stata translation (LLM-assisted)
│ └─ r_translation.R	R translation (LLM-assisted)
├─ LLM_conversation_log.txt	This conversation (source for this doc)
├─ PROJECT_PLAN.md	2-day plan with 12 steps
├─ README.md	Quick-start guide
└─ panel_replication_requirements.md	Original homework specification

Appendix: Instructions to Run the Code

```
# 1. Copy the project folder to any machine with Python installed

# 2. Install required packages (run once):
pip install pandas numpy matplotlib seaborn scipy statsmodels linearmodels openpyxl

# 3. Run the full workflow:
python scripts/main_workflow.py

# 4. All outputs appear in outputs/tables/ and outputs/figures/

# 5. To run a single step (e.g., only regressions):
python scripts/05_panel_estimations.py

# 6. To skip the dynamic panel step (Step 7), open main_workflow.py
#     and set skip=True for Step 7
```